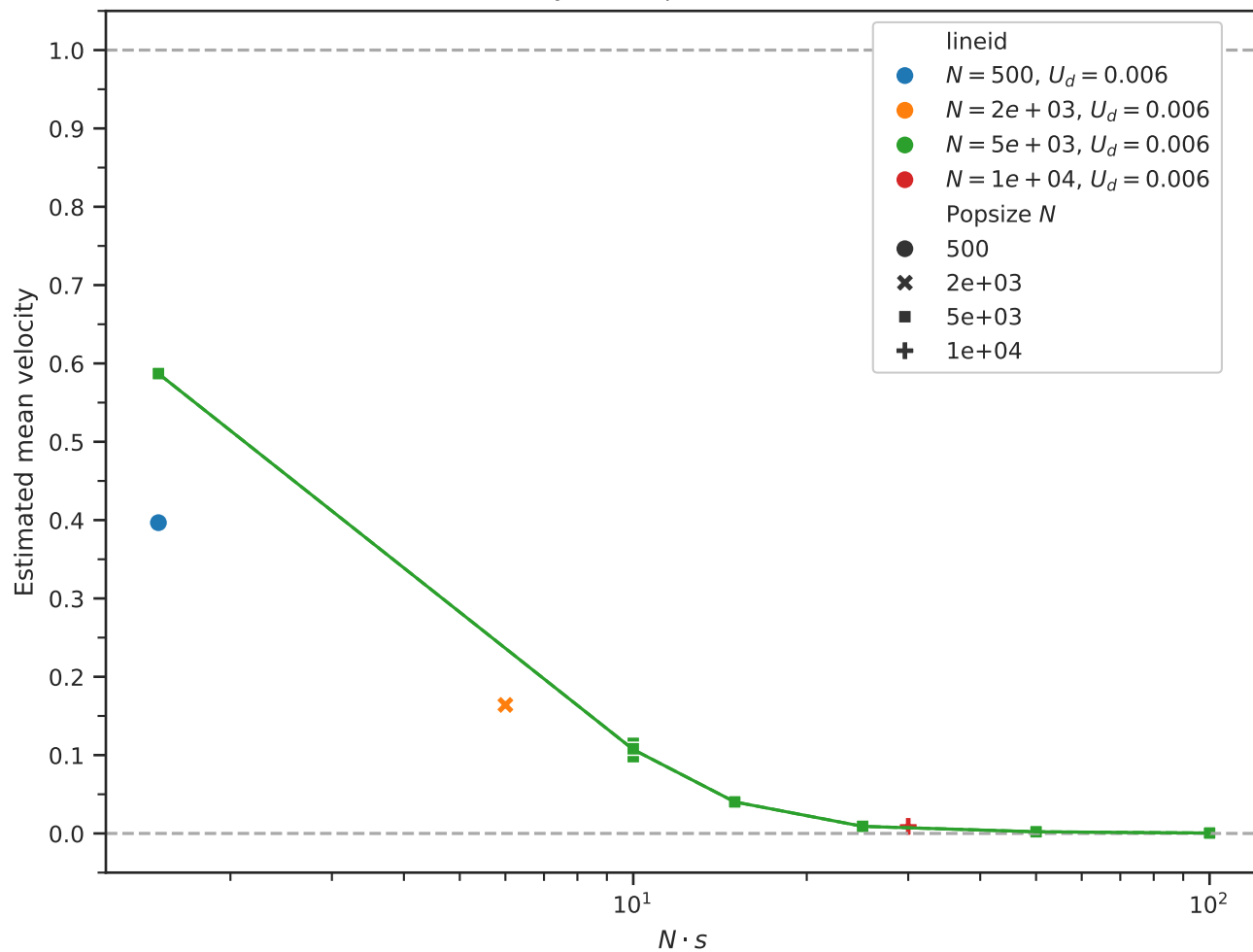
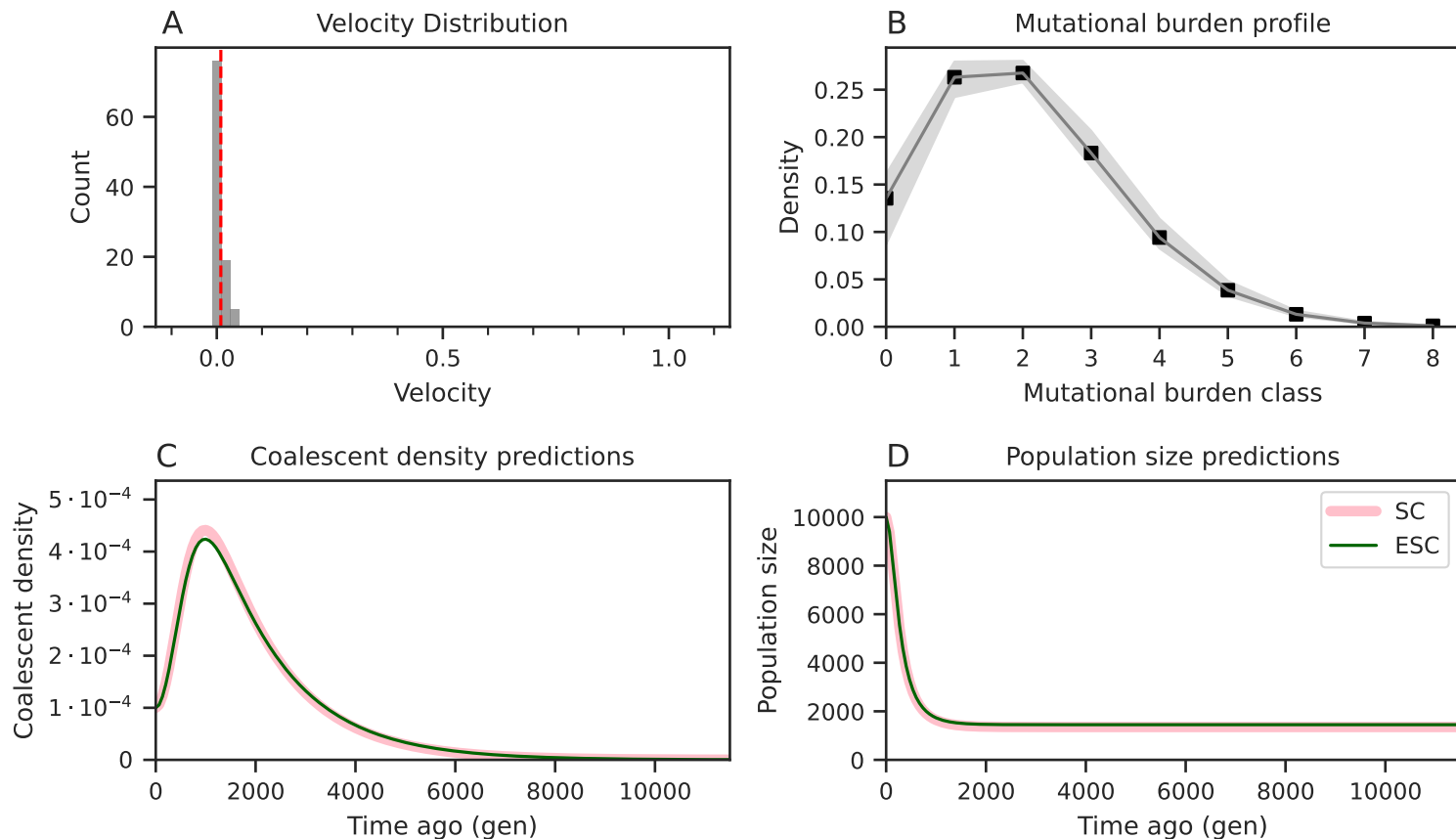


Mean velocity across parameter combinations

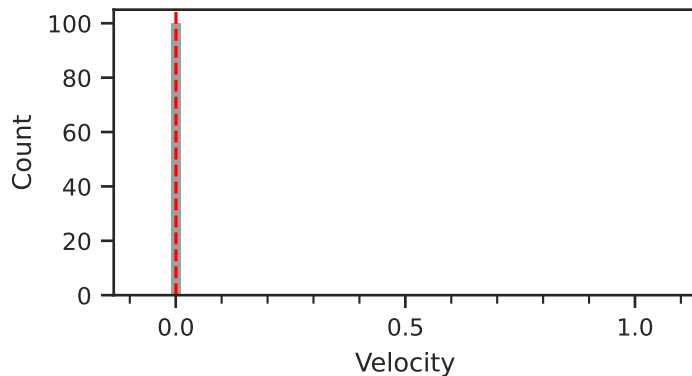


$N = 1e + 04$, $s = 0.003$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-4}$, $\phi = 4.06$
Mean Velocity= 0.0090 ± 0.0079 , $n = 100$ simulations

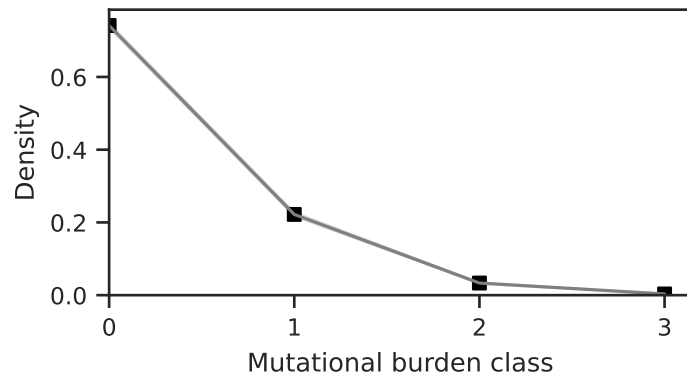


$N = 5e + 03$, $s = 0.02$, $U_d = 0.006$, $\sigma = 1 \cdot 10^{-4}$, $\phi = 74.1$
Mean Velocity= 0.0005 ± 0.0005 , $n = 100$ simulations

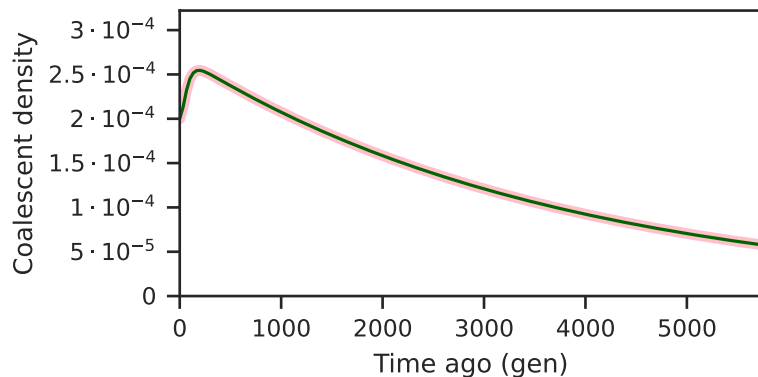
A Velocity Distribution



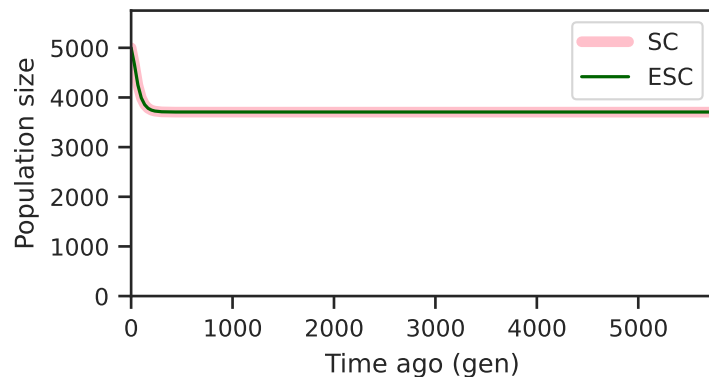
B Mutational burden profile



C Coalescent density predictions

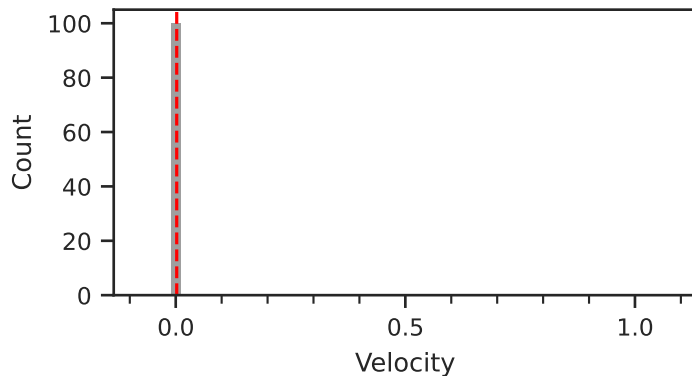


D Population size predictions

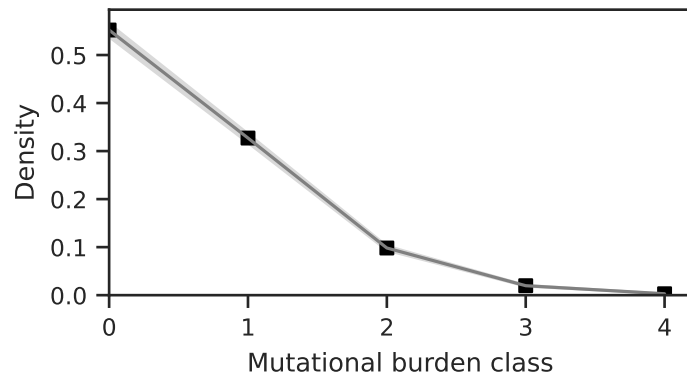


$N = 5e + 03$, $s = 0.01$, $U_d = 0.006$, $\sigma = 1 \cdot 10^{-4}$, $\phi = 27.4$
Mean Velocity= 0.0022 ± 0.0014 , $n = 100$ simulations

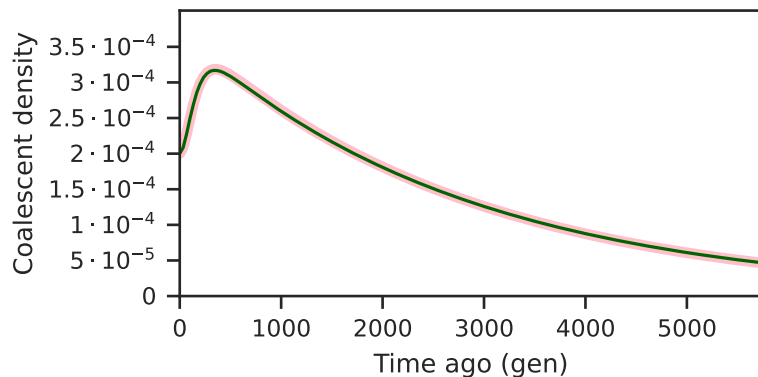
A Velocity Distribution



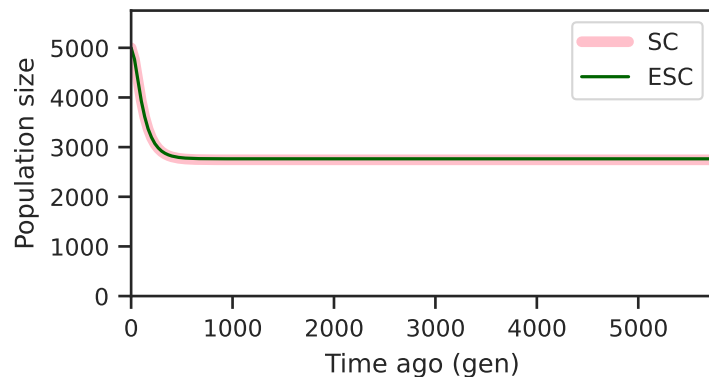
B Mutational burden profile



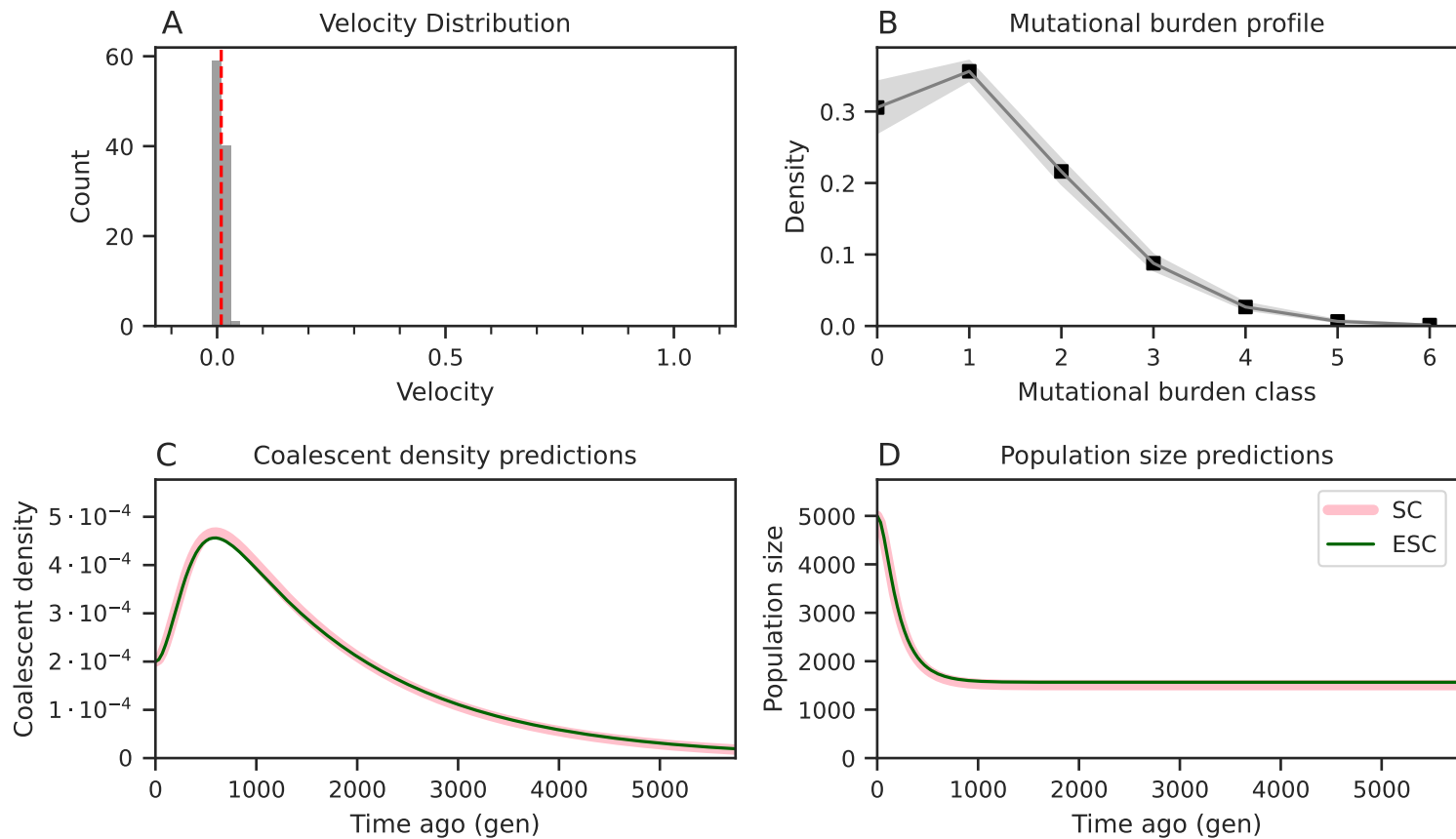
C Coalescent density predictions



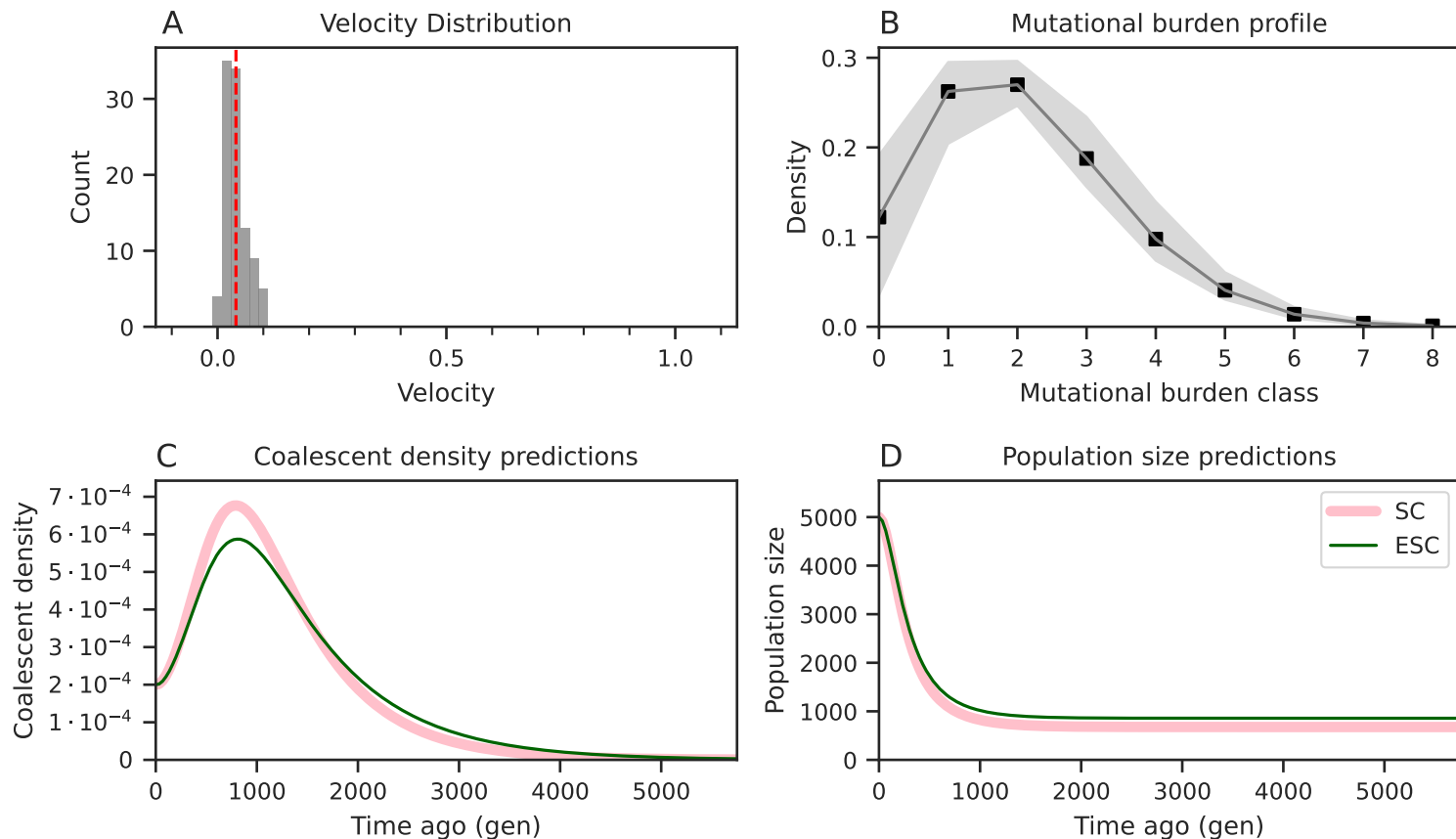
D Population size predictions



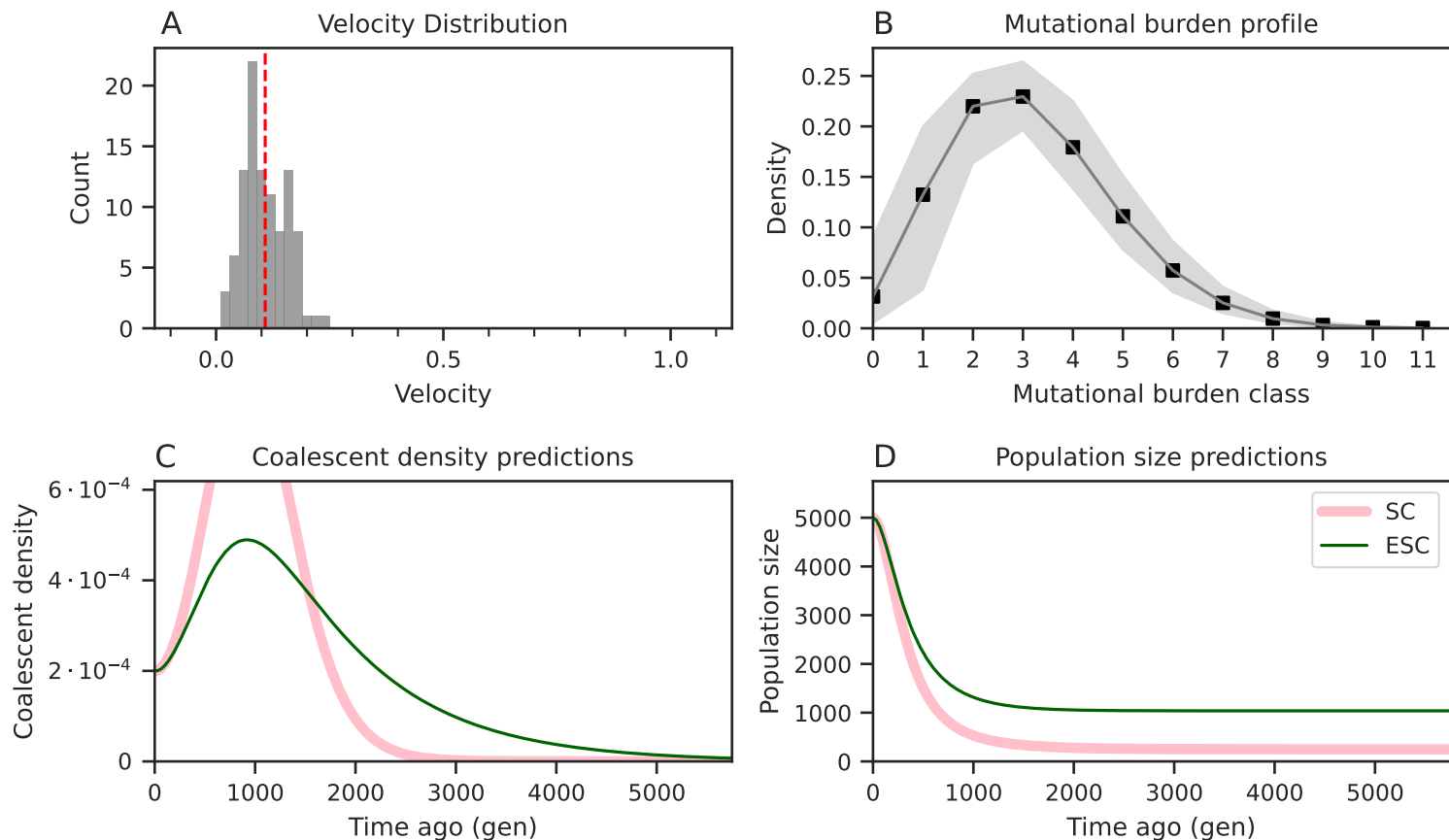
$N = 5e + 03$, $s = 0.005$, $U_d = 0.006$, $\sigma = 1 \cdot 10^{-4}$, $\phi = 7.53$
Mean Velocity= 0.0091 ± 0.0054 , $n = 100$ simulations



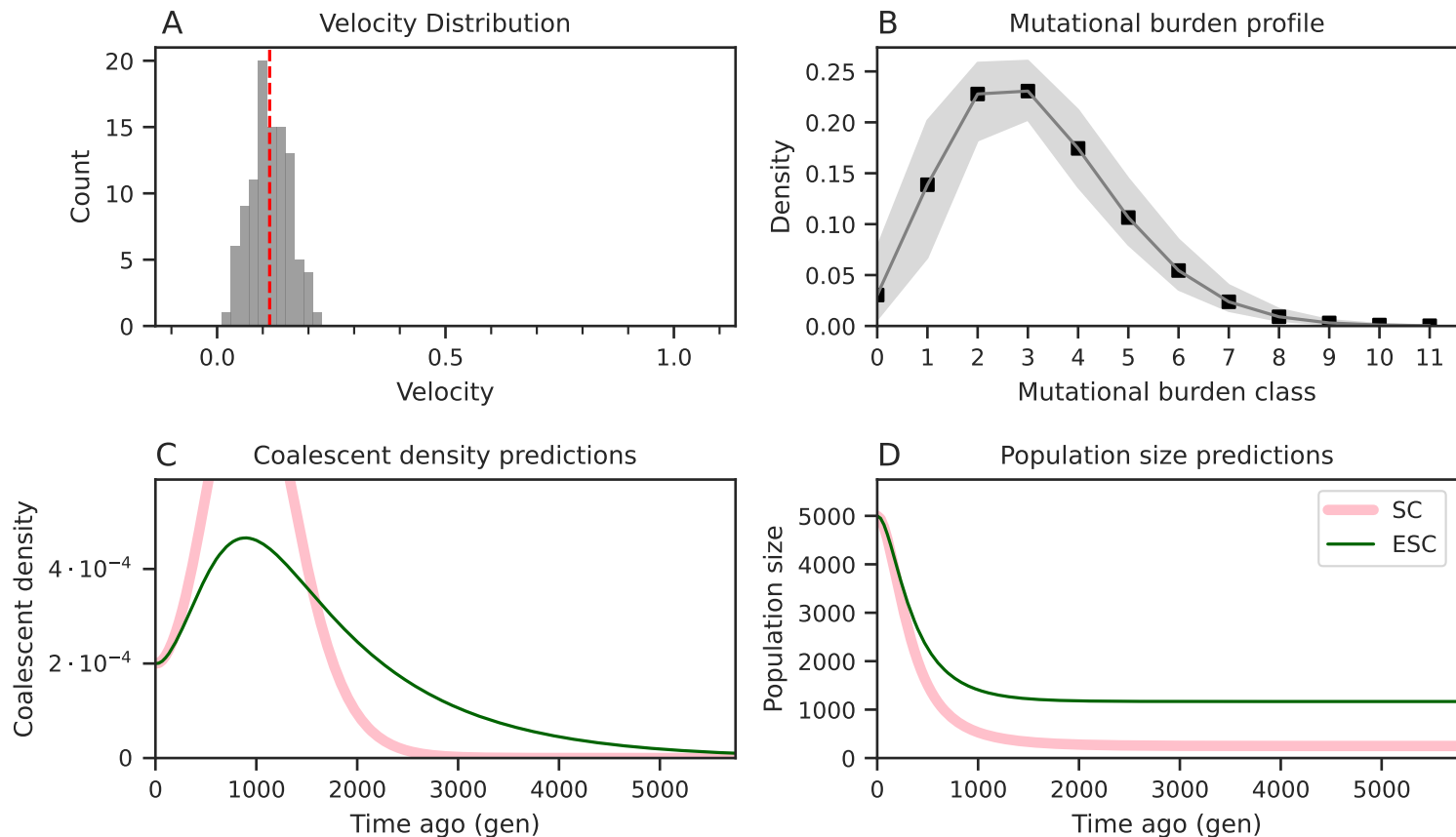
$N = 5e + 03$, $s = 0.003$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-4}$, $\phi = 2.03$
Mean Velocity= 0.0403 ± 0.0234 , $n = 100$ simulations



$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 0.001$, $\phi = 0.498$
Mean Velocity= 0.1081 ± 0.0469 , $n = 100$ simulations

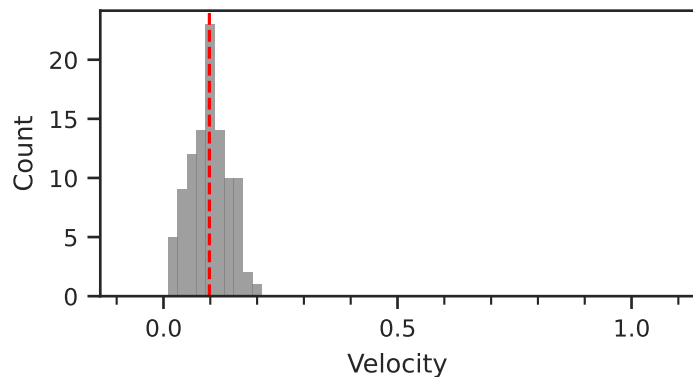


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-4}$, $\phi = 0.498$
Mean Velocity= 0.1150 ± 0.0418 , $n = 100$ simulations

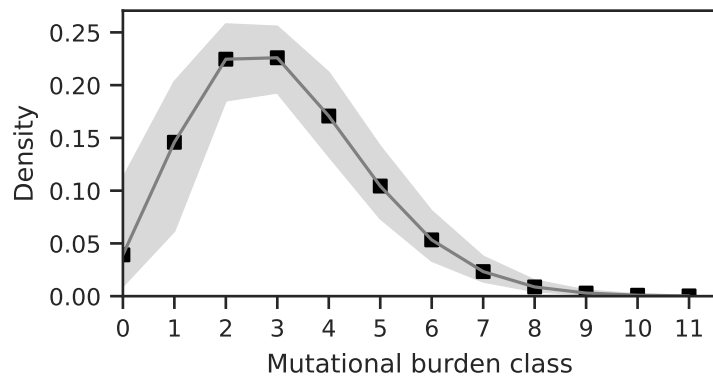


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 2 \cdot 10^{-4}$, $\phi = 0.498$
Mean Velocity= 0.0980 ± 0.0404 , $n = 100$ simulations

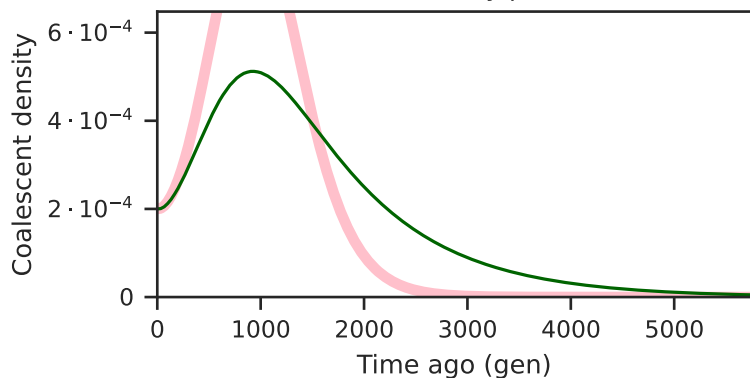
A Velocity Distribution



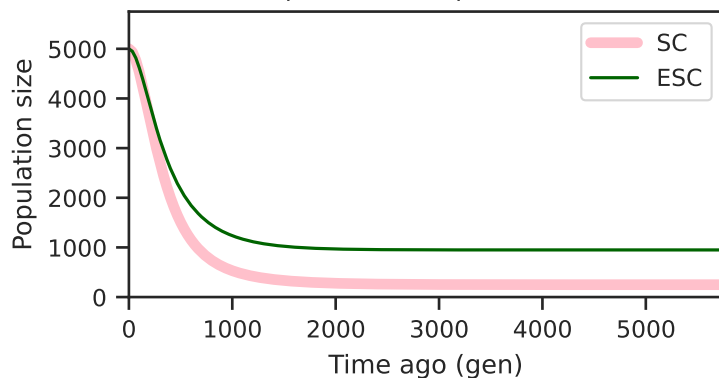
B Mutational burden profile



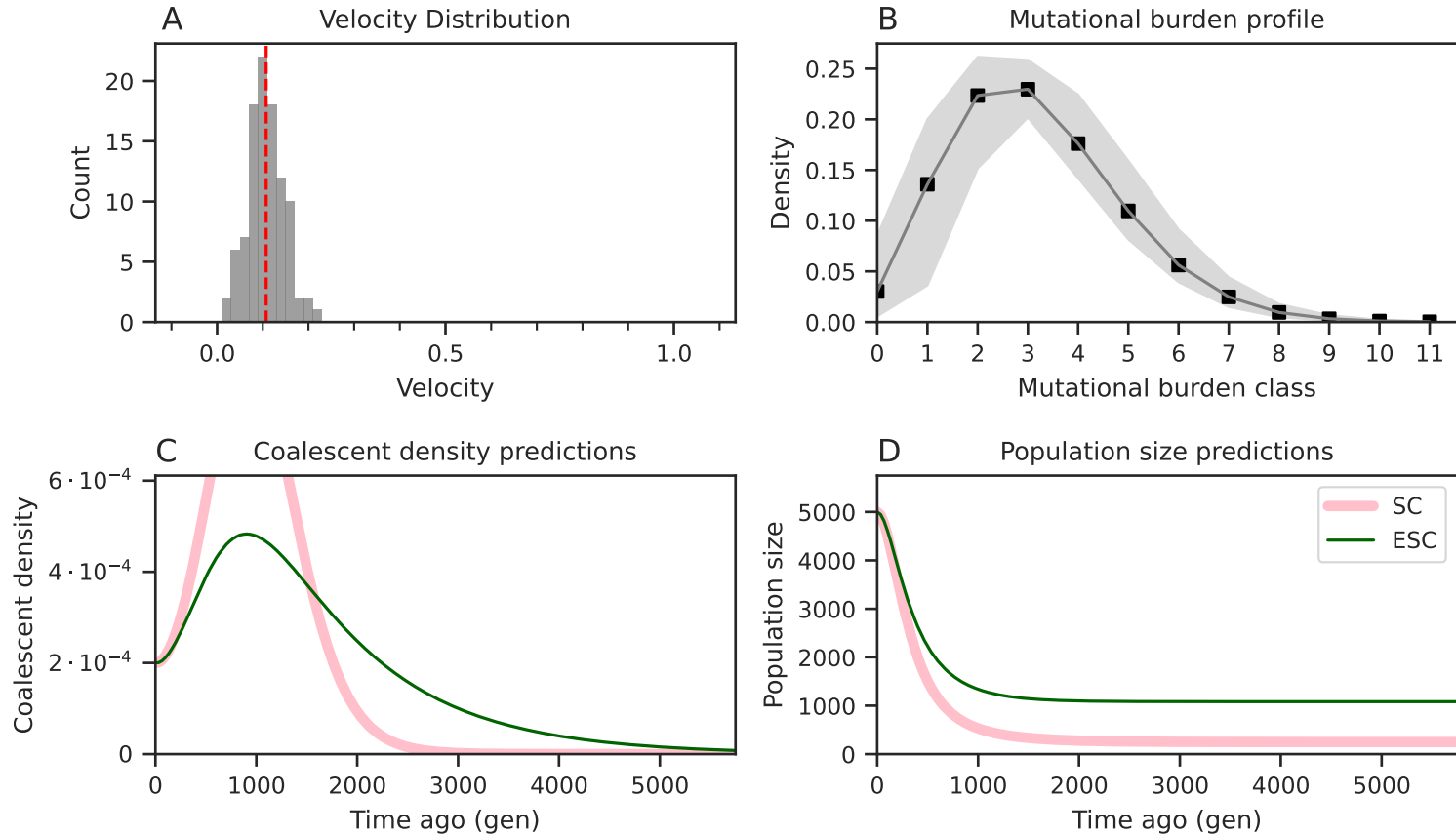
C Coalescent density predictions



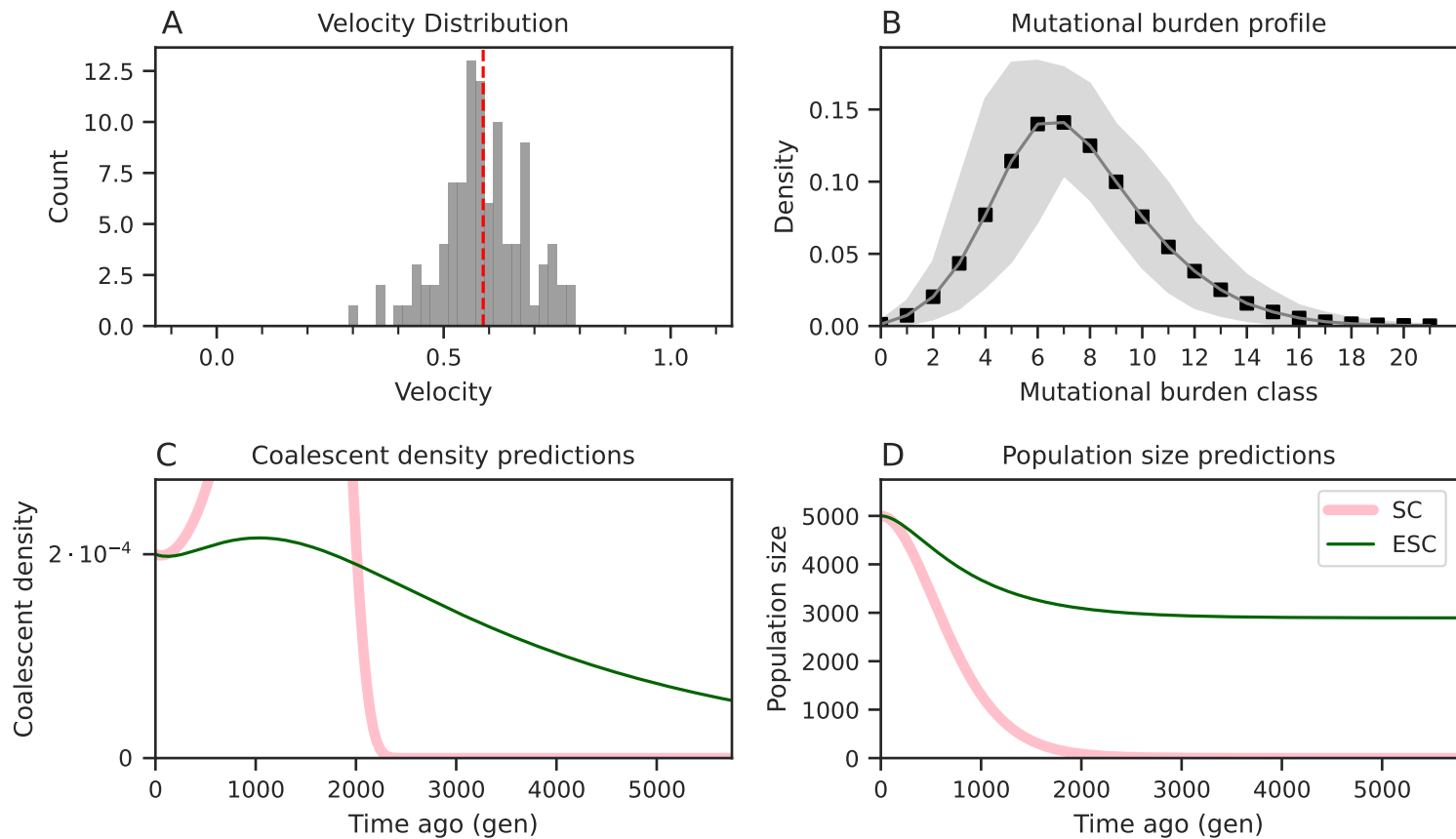
D Population size predictions



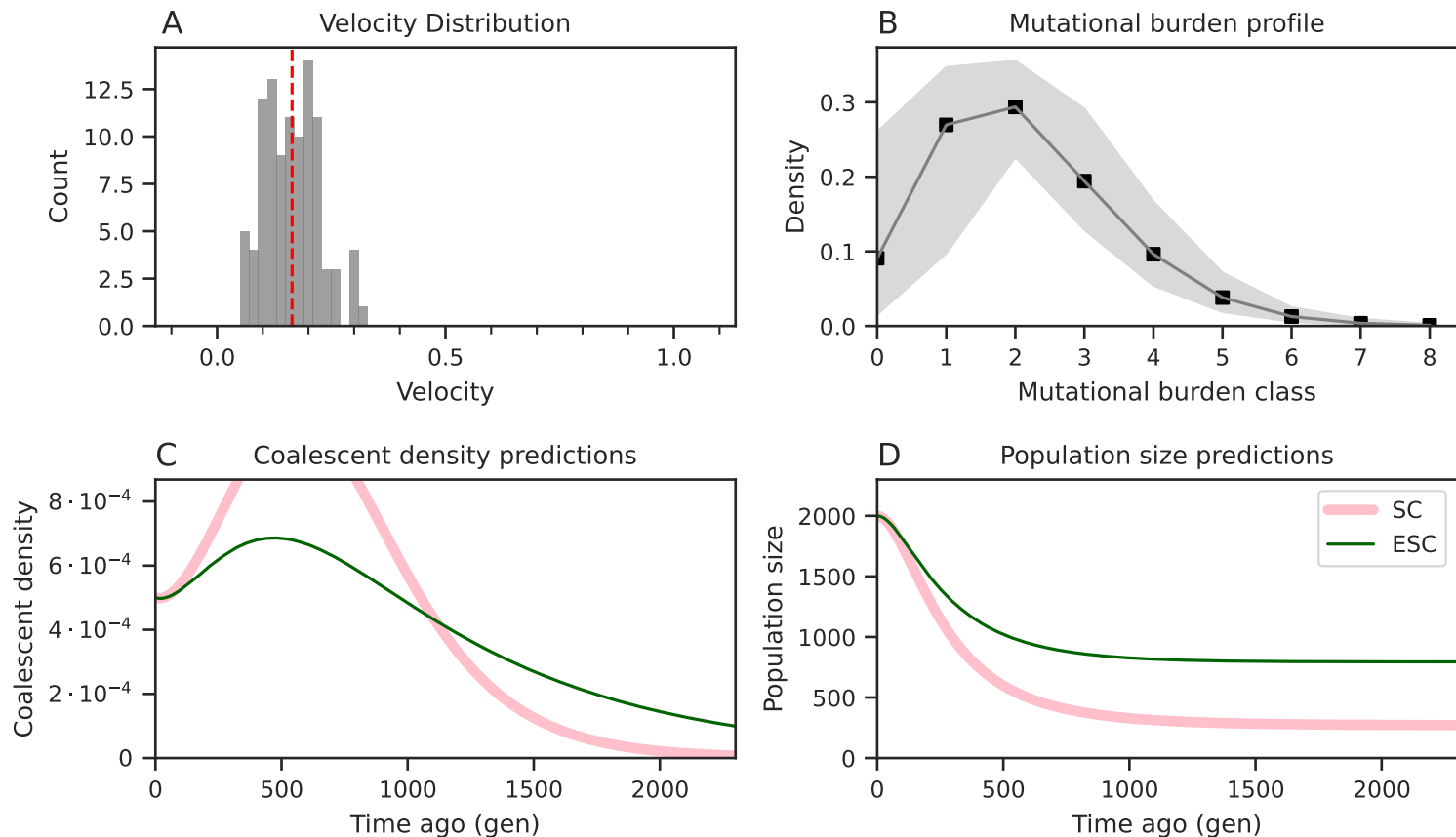
$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 1 \cdot 10^{-4}$, $\phi = 0.498$
Mean Velocity= 0.1074 ± 0.0387 , $n = 100$ simulations



$N = 5e + 03$, $s = 3 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 1 \cdot 10^{-4}$, $\phi = 3.09 \cdot 10^{-9}$
Mean Velocity = 0.5870 ± 0.0926 , $n = 100$ simulations



$N = 2e + 03$, $s = 0.003$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-4}$, $\phi = 0.812$
Mean Velocity = 0.1641 ± 0.0590 , $n = 100$ simulations



$N = 500$, $s = 0.003$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-4}$, $\phi = 0.203$
Mean Velocity = 0.3967 ± 0.1996 , $n = 100$ simulations

